Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 10
Hemodynamics of Systemic Circulation

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Recommended readings

Chapter 11: Hemodynamics. Medical Physiology: Principles for Clinical Medicine, 6e.

Rodney A. Rhoades, David R. Bell.

Klabunde Cardiovascular physiology concepts. 2nd ed. Lippincott. Chapter 5: pp 85, 93-108

Review: DLA and Lecture: Introduction to the cardiovascular system

- This lecture discusses the physical laws that govern the flow of blood in the blood vessels.

Slides with blue background are the supplemental information associated with the previous slides and thus, might not be explained during the lecture time. However, please read the slides for clarification as they are equally important to know.

Learning Objectives

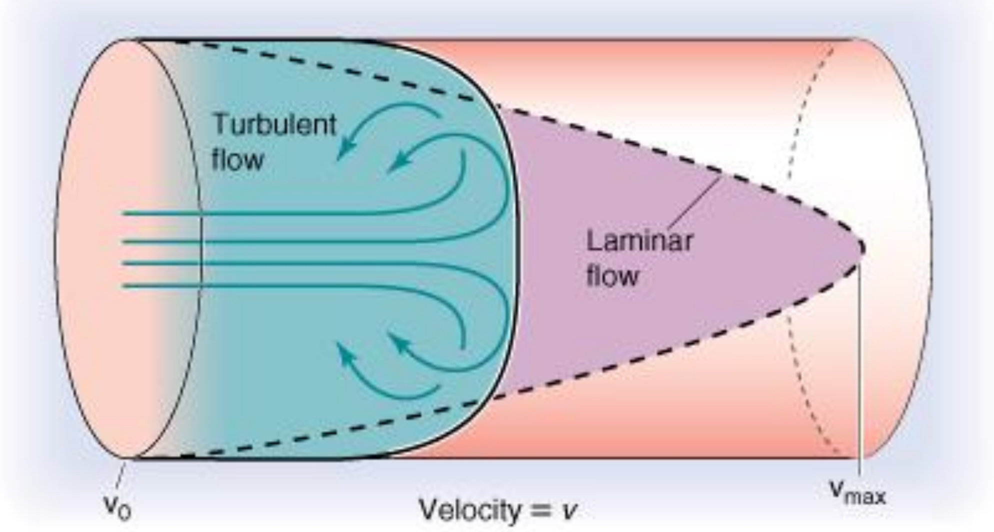
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0113** List the factors that shift laminar flow to turbulent flow. Compare the relationship between velocity, viscosity, and audible events, such as murmurs and bruits.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0114**Describe single line blood flow and relate this to sickle cell anemia
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0115** Describe the relationship between wall tension, pressure and radius: La Place's law
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0116** Given systolic and diastolic blood pressures, calculate the pulse pressure and the mean arterial pressure.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0117** Describe how arterial systolic, diastolic, mean, and pulse pressure are affected by changes in a) stroke volume, b) heart rate, c) arterial compliance, and d) total peripheral resistance.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0118** Describe how gravity affects blood flow and relate it to blood pressure gradient
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0119** Describe factors that affect venous return
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0120** Using Fick's Principle, discuss the dye dilution and thermodilation methods to calculate the cardiac output.

Lecture Outline

- **Types of blood flow**
  - **Laminar vs turbulent flow,**
  - **Reynolds equation,**
  - **Single line flow**
- Physiological properties of vessels
  - La Place's Law of wall tension
  - Compliance vs elastance
- Pulse Pressure and Mean Arterial Pressure
- Effect of gravity on blood pressure
- Factors affecting venous return
- Measurement of Cardiac Output: Fick's principle

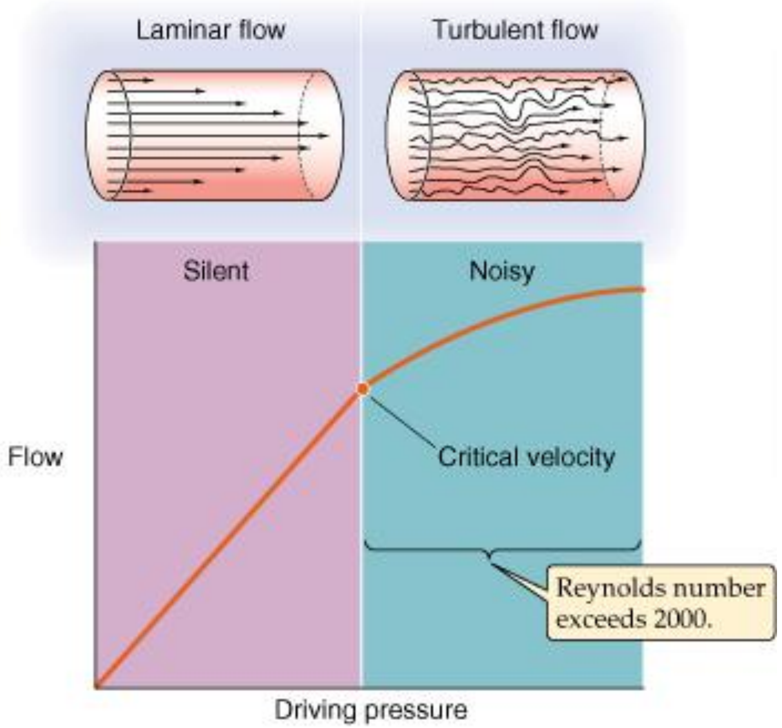
TYPES OF BLOOD FLOW: Turbulent and Laminar

Turbulent flow	Laminar flow
• NOISY – heard by auscultation (Bruit). • Fluid particles move randomly in all directions, Eddies are chaotic. • Requires an increased pressure to maintain flow (flow is not as efficient) • Occurs at large branch points, in diseased or narrowed arteries, across stenotic heart valves, and ventricles	• Silent • Parabolic profile - Concentric rings of equal flow rates; • Slowest at the edges (friction with the walls), • Highest velocity at the centre • **Normal** – linear, smooth flow on blood vessels



Reynold's equation

When does laminar flow become turbulent flow?



Boron and Boulpaep

When the Reynold's number is exceeded > 2000

– laminar flow becomes turbulent .

Factors which ↑ Turbulence **pro-turbulence factors**	Factors which ↓ Turbulence **anti – turbulence factors**
High flow velocity (v)	↑ Viscosity (η)
Large vessel diameter (D)	
High blood density (ρ)	

$$Re = v\,D\,\rho\,/\,\eta$$

Reynold's equation

When does laminar flow become turbulent flow?

$$Re = v\ D\ \rho\ /\ \eta$$

Factors which ↑ Turbulence pro-turbulence factors	Factors which ↓ Turbulence anti – turbulence factors
High flow velocity (v)	↑ Viscosity (η)
Large vessel diameter (D)	
High blood density (ρ)	

Apparent contradiction?

1. **High velocity flow favors turbulence**, high velocity flow is found in narrow diameter vessels – greatest effect on promoting turbulent flow

e.g. stenosis increases turbulence

2.<u>Large vessel diameter is associated with increased turbulence</u>, - Increases velocity of flow..

This may appear paradoxical because increasing **r** increases the cross-sectional area

This might be expected to decrease the velocity (if Q remains unchanged)

But – based on Poiseuille – flow increases with increased radius, cross sectional area increase – therefore velocity would increase ,

Blood flow is slower at the surface of the vessels: the larger the vessel the smaller the surface area: volume ratio. Contact with the vessel wall (surface area contact) slows velocity (v) because of resistance. Decreased SA/volume ratio occurs with ↑ diameter (D), Reduced blood in contact with a surface area, Increased turbulence.

-Think a bigger tube - more area in the middle for turbulence to occur.

TYPES OF BLOOD FLOW: single line flow

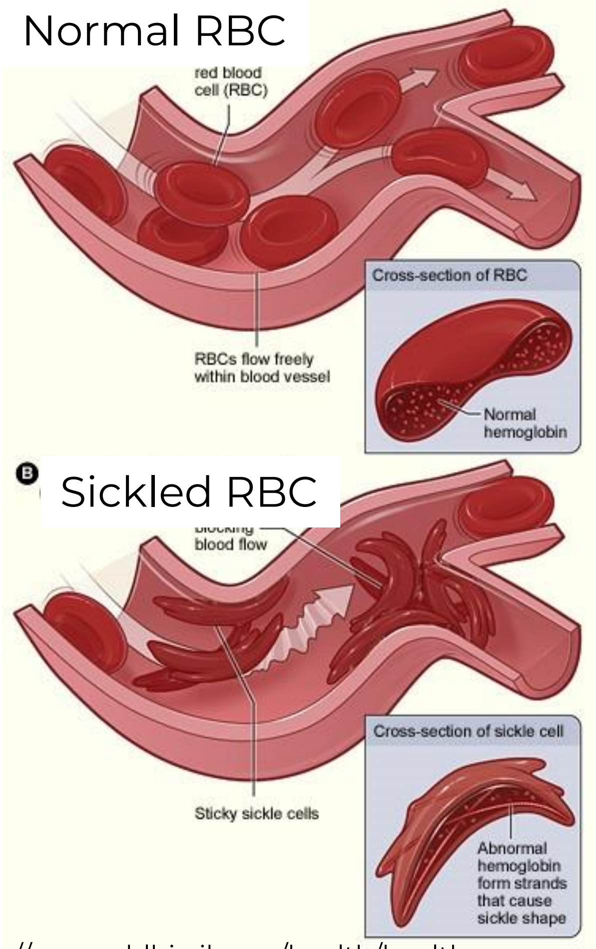
Single line flow

- Occurs in capillaries
- RBC diameter > capillary diameter
- RBC flexes as passes through the capillary
- RBC actually flows faster than plasma (less friction in axial flow, vs plasma at boundary flow)

- Clinical application:

Sickle cell anemia: Red blood cell (RBC) is rigid, **Hemogobin** Hb rod-like and RBC sickle shape. Do not pass easily through capillary → tissue ischemia and painful "sickling crisis".

Low blood flow: RBCs may stick together and to endothelial lining.



http://www.nhlbi.nih.gov/health/health-topics/topics/sca

Lecture Outline

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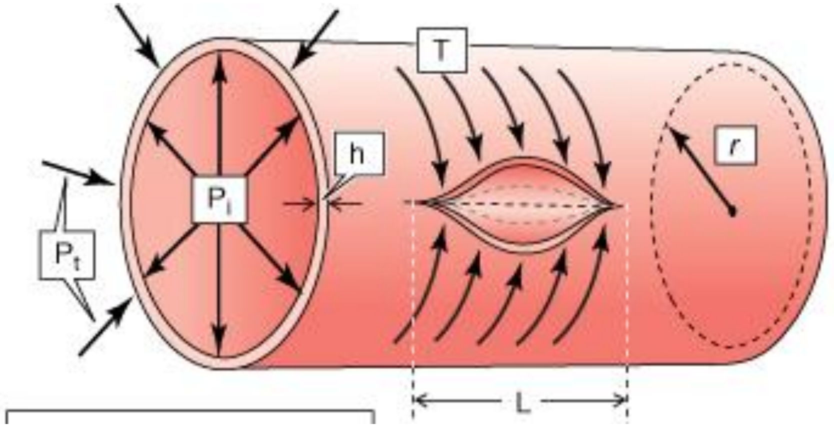
La Place’s Law of Wall Tension



<u>LaPlace’s Law for a cylinder:</u>

<u>T=</u> ΔP*r

Tension (T)
Transmural pressure (ΔP)
Radius (r)

Laplace’s law relates the tension in an arterial or venous wall with the pressure that the elastic tube can apply to material inside the tube.

Wall tension counteracts transmural pressure
Transmural pressure (ΔP) = the difference between Pi-Pt, ΔP = Pi - Pt

- Pi – pressure inside trying to distend the vessel wall
- Pt – pressure outside trying to compress it

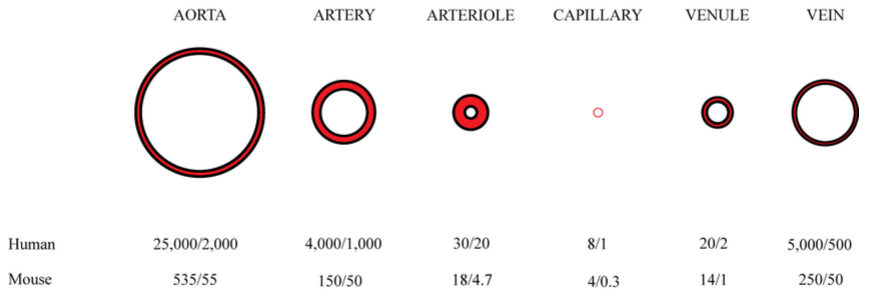
Wall tension changes with radius.
If the pressure inside the vessel is constant, the larger the radius (diameter) of the vessel, the greater the wall tension needed.

Wall tension is reduced by wall thickness (h), If the pressure inside the vessel is constant

Application of La Place’s Law of Wall Tension

LaPlace’s Law: T= ΔP*r wall tension is proportional to the vessel radius for a given blood pressure

The closed circulatory system is composed of arteries, veins, and capillaries.
Arteries are thick walled, resistance vessels and carry blood away from the heart.
Veins are capacitance vessels and carry blood to the heart.
Capillaries are small tubes that exchanges substances by connecting arterioles (small arteries) and venules (small veins).

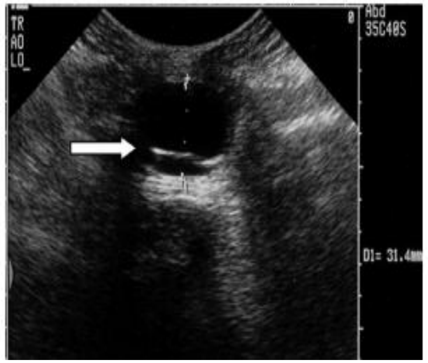


How is tension in the walls of blood vessels affected by their size(radius) and thickness?

- For a given blood pressure, increasing the radius of the cylinder leads to a linear increase in tension.
- If driving pressure is increased, wall tension also increases.

The implication of this law:
Larger arteries have large radius – wall tension is increased.
Large arteries withstand high blood pressures - must have stronger walls to withstand twice the wall tension.
Capillaries also carry high blood pressure, but unlike arteries, capillary walls are thin. Their small size leads to a reduced level of tension so that thick walls are not necessary.

What happens when the wall tension is too great?



https://radiologykey.com/34-abdominal-aortic-aneurysm/
https://www.ejves.com/article/S1078-5884%2820%2930523-2/fulltext

First, wall tension is proportional to vessel radius, according to Laplace's law: $T = P \times r$, where T is circumferential wall tension, P is transmural pressure, and r is mean vessel radius.

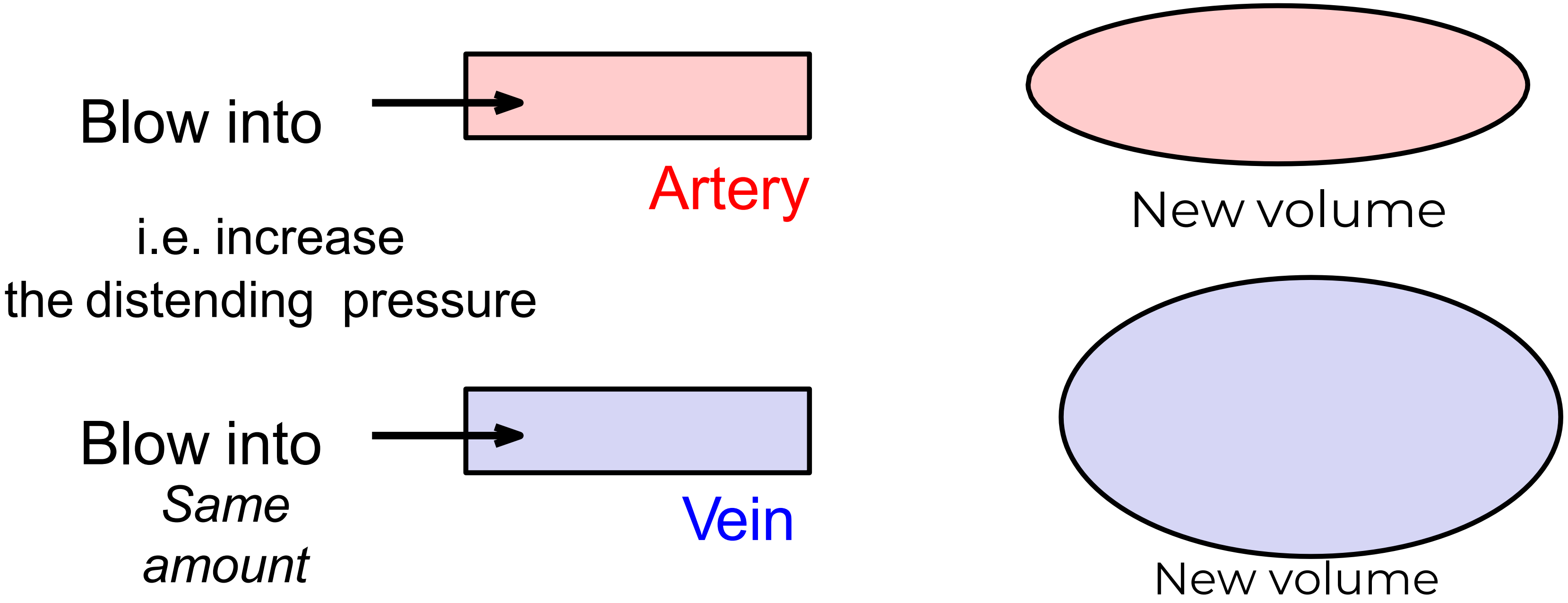
Second, increased tension stress from blood pressure results in progressive vessel dilatation and weakening of aortic media, which lead to enlargement of aortic aneurysm.

Third, when mechanical stress on wall exceeds strength of wall tissue, aortic aneurysm ruptures.

Complications of uncontrolled Hypertension

Compliance and elastance
Comparing the response in arteries and veins to distending pressures

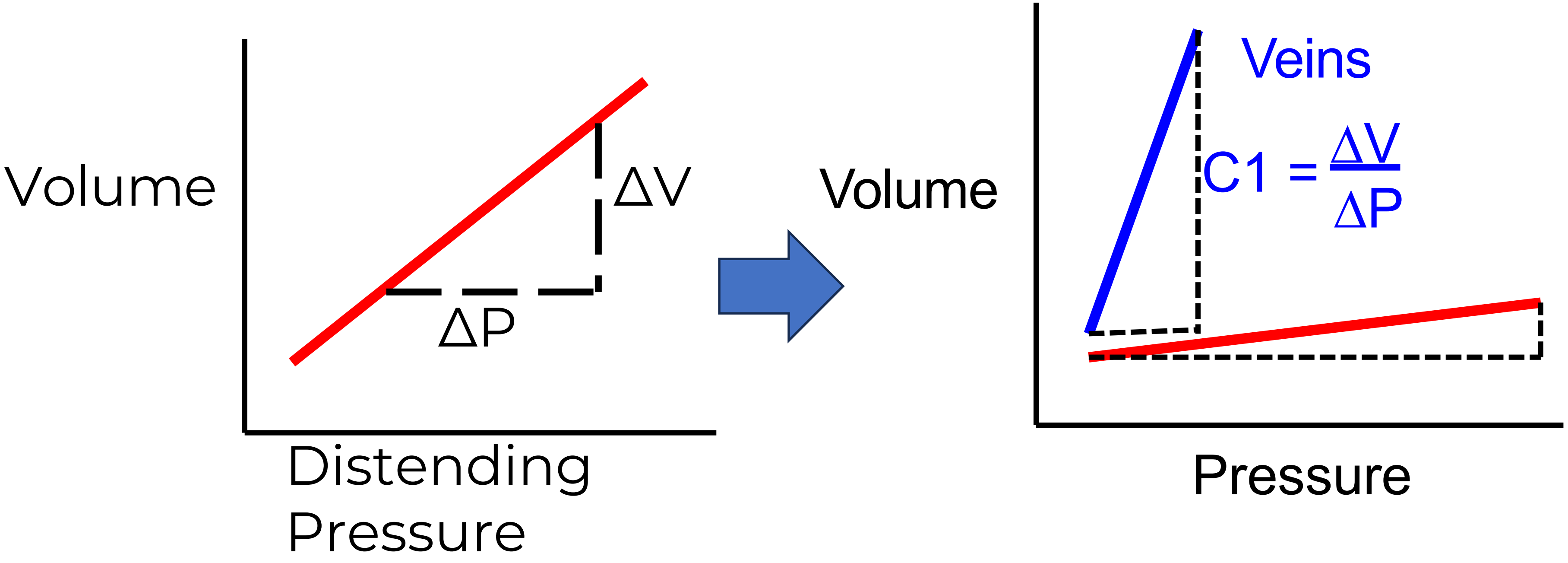
Large arteries and veins contribute very little to overall resistance to blood flow. Therefore, changes in their diameter have minimal effect on blood flow.
The compliance and elastic properties of arteries and veins are important because it determines how much pressure the vessel can withstand or how much blood can be stored in them. Veins, especially, act as reservoirs of blood.



Compliance
Comparing the response in arteries and veins to distending pressures

The term compliance is used to describe how easily a chamber of the heart, or the lumen of a blood vessel expands when it is filled with a volume of blood.(change in the shape of a structure when the mechanical load is applied)

Compliance (C) is defined as the change in volume (ΔV) divided by the change in pressure (ΔP). C = ΔV / ΔP

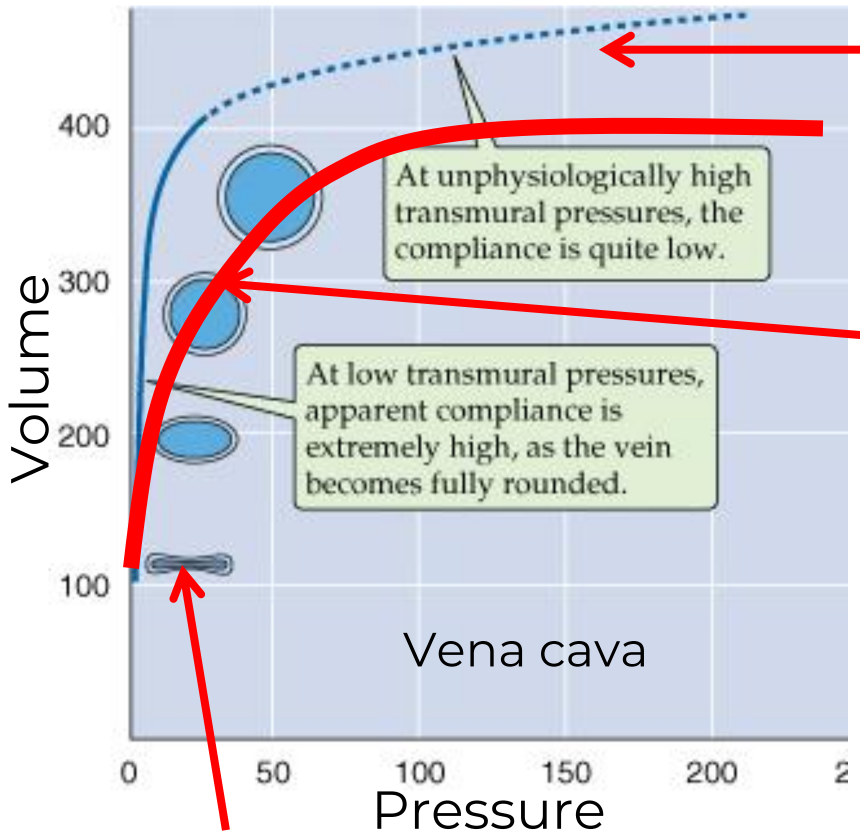


Compliance is a measure of “distensibility/ stiffness/capacitance”

Veins are more distensible than arteries.

Small change in pressure – large increase in volume

Veins - compliance and elastic properties



At very high pressures- all fibers stretched,

Vascular tone (SNS) shifts compliance curve down and to the right.

At low pressures- veins collapsed!

VEINS ARE

VERY COMPLIANT

- At low press/vol, large veins collapse. venous pressure not so high.
- Small changes in pressure with large changes in volume. Steep slope at low pressures:
- Inc press/vol, inc circular shape
- Until vessels attain circ shape, walls not stretched much
- once the vein has a circular profile it does not increase its diameter because it has very little elastin in its walls)
- Vena cava: capacitance vessels, adjustable reservoirs of blood.

Boron and Boulpaep

What happens when veins constrict or dilate?

↑Sympathetic → Venoconstriction; ↓ *venous compliance*
↓Sympathetic → Venodilation; ↑ *venous compliance*

<u>Effects of venoconstriction</u>

Venoconstriction "squeezes" blood out of the veins – and shifts it towards the heart. i.e. ↑ VR Note: venoconstriction is akin to ↓

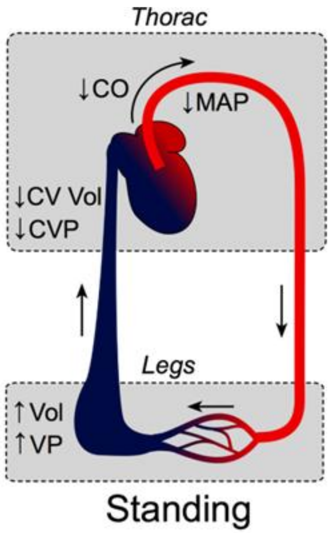
<u>Effects of venodilation</u>

Conversely, venodilation allows more blood to pool in the veins and ↓ VR. Note: venodilation is akin to ↑ compliance

<u>*Venodilation and venous pooling (from other causes)</u>

1. "Back-up" – reduced venous return can increase blood volume in venous system. If the distending pressure is increased, then veins distend to accommodate blood.
2. "Gravity" - Standing increases the distending pressure in the veins of the foo. Compliant veins accommodate a large volume of blood.

Venous pooling decreases venous return and decreased venous return causes venous pooling



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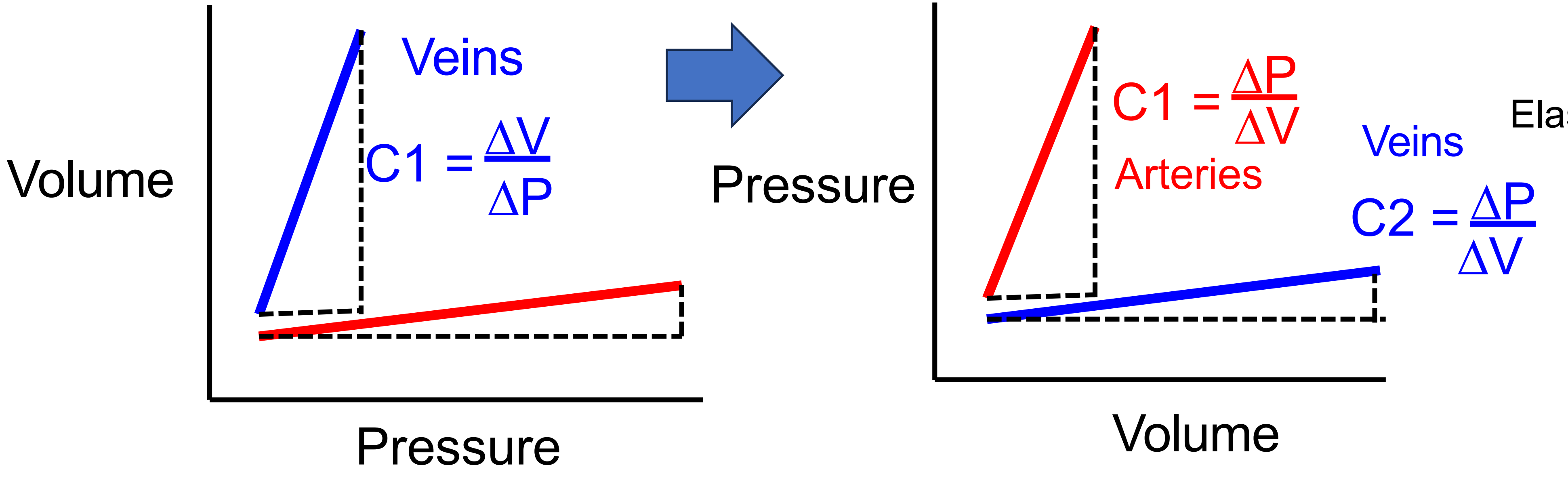
Elastance

Comparing the response in arteries and veins to distending pressures

Elastance is a term used to describe the elastic recoil of blood vessel when it is filled with a volume of blood/distending pressure. (resistance to change the shape when mechanical load applied)

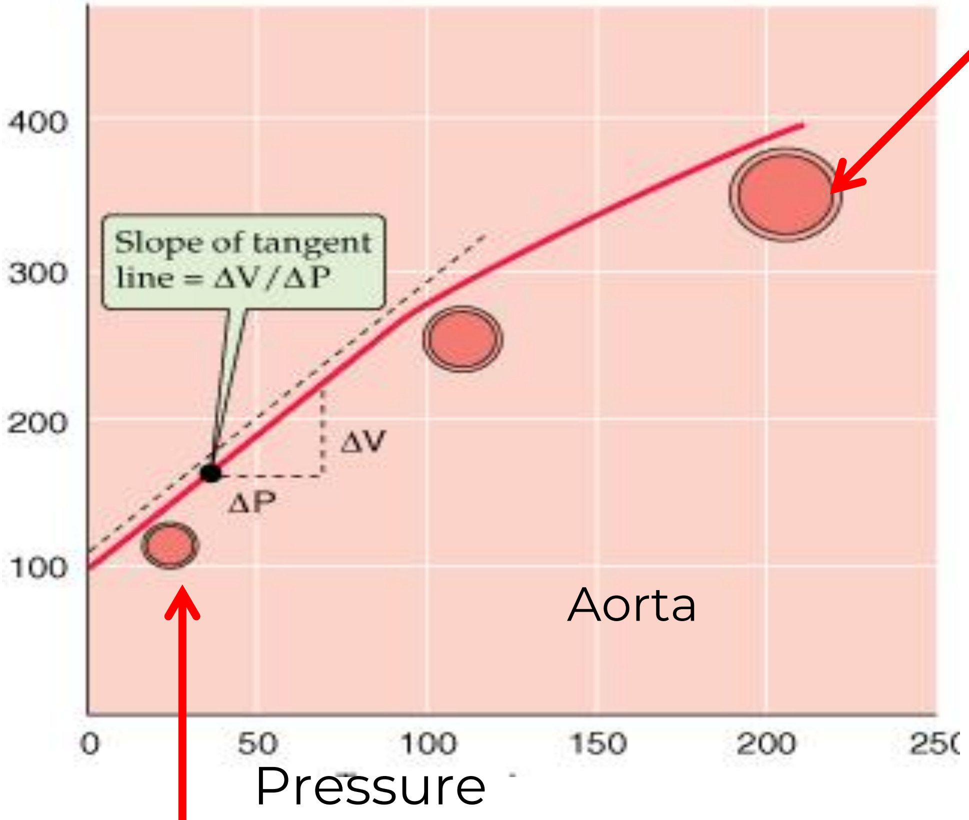
Elastance *(E) is defined as the change in pressure (ΔP) divided by the change in volume (ΔV)* . E = ΔP / ΔV

Arteries are more "elastic" than veins



Elastance = Slope = $\frac{1}{\text{Compliance}}$

Aorta- compliance and elastic properties



At low pressures – arteries are patent!

At very high pressures – arteries are also patent!

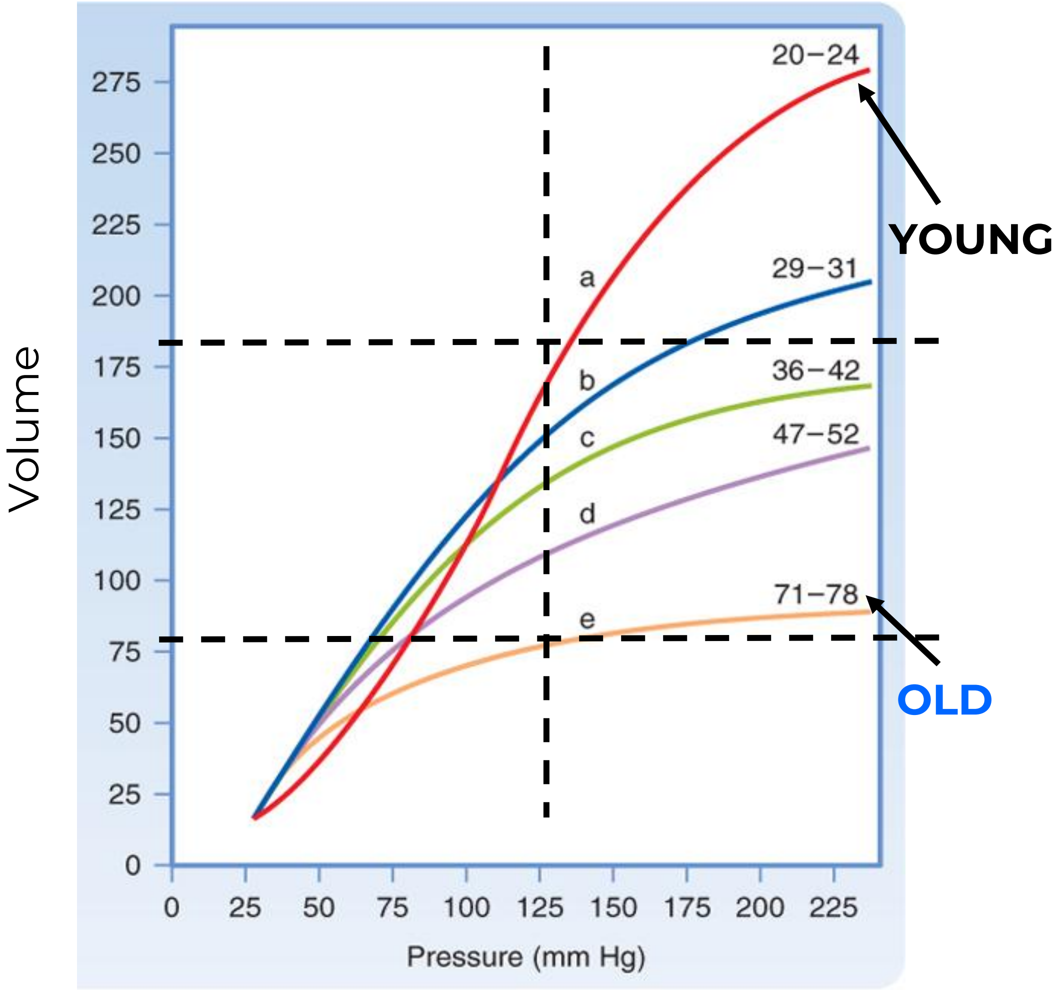
- In early phase of cardiac ejection, the aortic volume increases because there is more blood entering the aorta than leaving it. The elastic aorta is stretched.
- Towards the end of systole and during diastole, the stretched aortic and arterial walls recoil and in the process give up their stored potential energy. This reconverted energy drives the blood around the circulation.
- Aorta (and arteries) develop and withstand high pressures. Stable resistance. Therefore, changes in their diameter have minimal effect on blood flow. They convert the pulsatile flow from the heart into steady flow through the vascular beds.

- If compliance is decreased in arteries – stiffness causes an increased transmural pressure.

ARTERIES ARE VERY ELASTIC

Boron and Boulpaep

Does compliance in the arteries change with age?



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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Aortic "compliance" is clinically important in the elderly. "Old" artery less compliant/ stiffer than "young" artery because of loss of elastin.

At a given MAP, "old" arteries hold less blood than "young" arteries. E.g. at 125 mmHg, 75 vs 175 mls.

<u>For an "old" artery to hold the same volume of blood as a "young" artery, the pressure in the "old" artery must be higher.</u>

E.g. Ejecting 75 ml can be done easily at low pressures in a young artery. ~75 mmHg
However, 75 ml in an older artery requires a larger pressure ~125 mmHg.

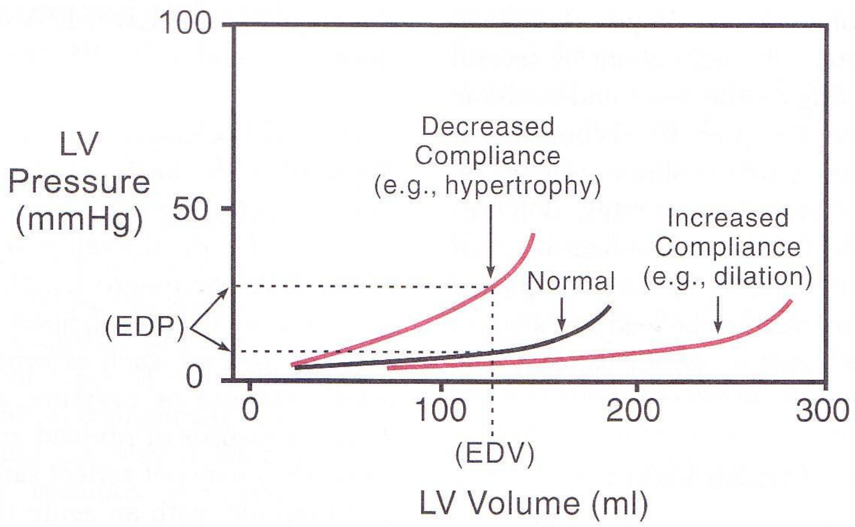
Diastolic BP may be normal (at ~ 80 mmHg)

High systolic BPs and increased pulse pressure

= "Systolic Hypertension of the Elderly"

Should it be managed with drugs ??

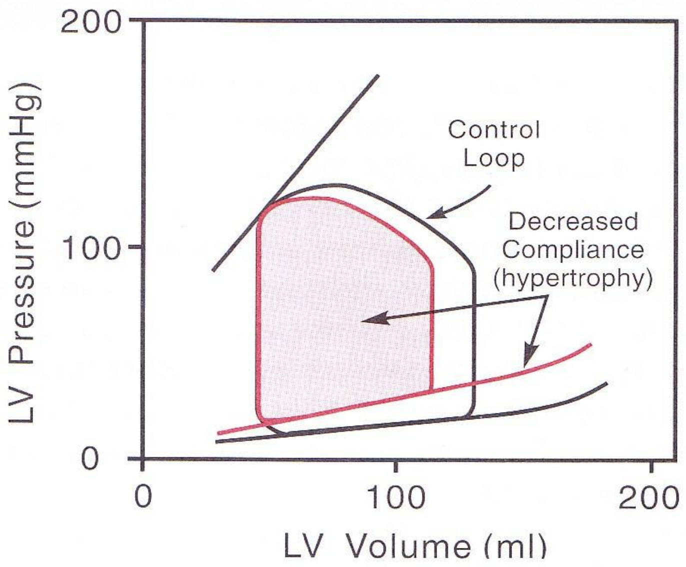
A quick look back at compliance in ventricles



- ↓ ventricular compliance (hypertrophy)

  = ↑LVEDPr, ↓ LVEDV

- ↑ ventricular compliance (dilated cardiomyopathy)
  = ↓LVEDPr, ↑ LVEDV



- LV hypertrophy
- ↑ wall thickness
- ↓compliance
- ↑ LV End Diastolic Pr
- ↓ Filling
- ↓ Stroke volume
- ↓ Cardiac output

Cardiovascular Physiology Concepts: Klabunde LWW

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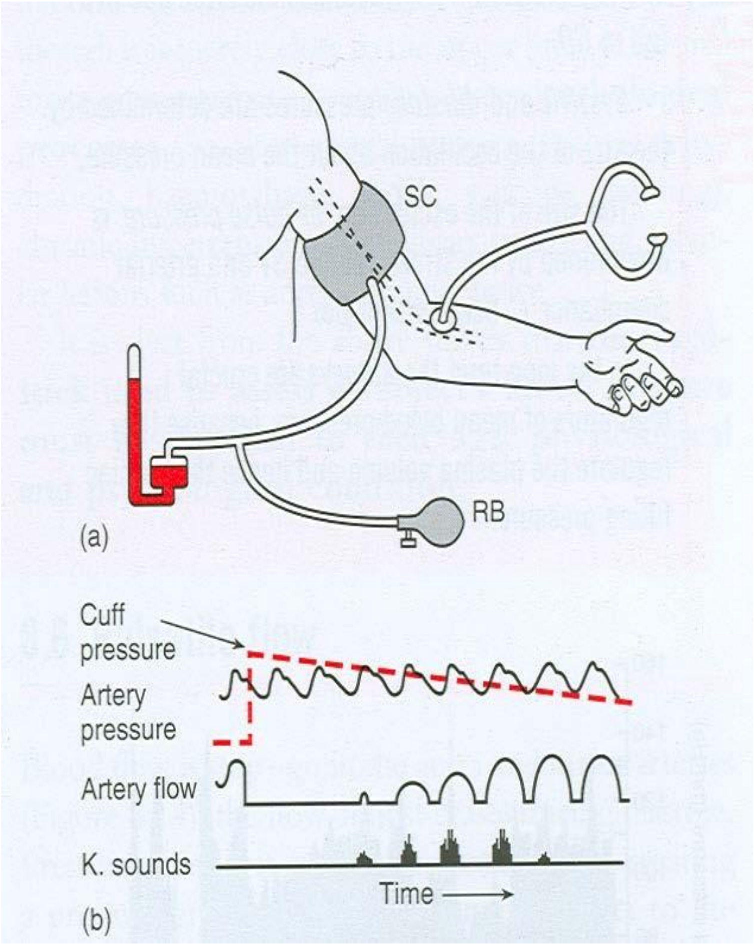
Measurement of Blood Pressure

Sphygmomanometry



Levick

Cuff pressure:

Cuff inflation occurs until pressure in cuff is greater than arterial pressure, obstruction of blood flow in artery. Cuff deflation occurs slowly until cuff pressure is less than arterial blood pressure (SBP/DBP)

Arterial pressure wave/pulse wave

Apex heartbeat is heard simultaneously while feeling pulse because of arterial pressure wave. When blood is ejected at high pressure into the aorta – a **pressure wave** is set up which is transmitted along the **walls of the arteries**. Shape of the pressure wave becomes narrower and taller as it proceeds down the arterial tree. Arterial pulsations causes oscillations.

Korotkoff sounds:

On auscultation: 1st Korotkoff sound occurs when cuff pressure is just below systolic BP because of transient spurt of blood into artery (turbulent flow); Korotkoff sounds grow louder then quiet; Korotkoff sounds disappear as the cuff pressure approaches diastolic BP – artery is almost fully patent (laminar flow)

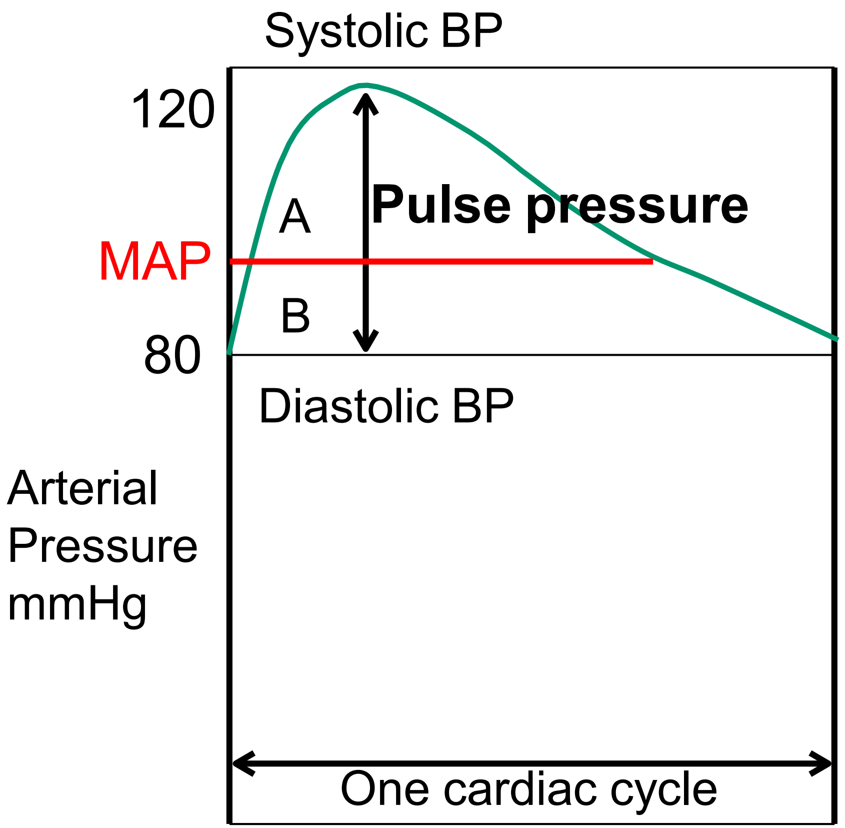
- Blood pressure reading consists of systolic and diastolic blood pressure readings.
- A mean arterial pressure (MAP) and pulse pressure (PP) can be calculated from SBP and DBP recordings.

Physiological Factors affecting BP

Age:	6 yr	120/65 mmHg
	30 yr	125/80 mmHg
	70 yr	180/90 mmHg
Sleep	80/50 mmHg	
Exercise	Dynamic	↑ by 10 – 40 mm Hg
	Isometric	Can ↑ by 100 mm Hg
Emotion/Stress		↑ BP
Pregnancy		BP falls

BP of 130/90 mmHg in a 6-month pregnant woman is serious

Arterial pressure wave:
Blood Pressure in the Aorta during one cardiac cycle

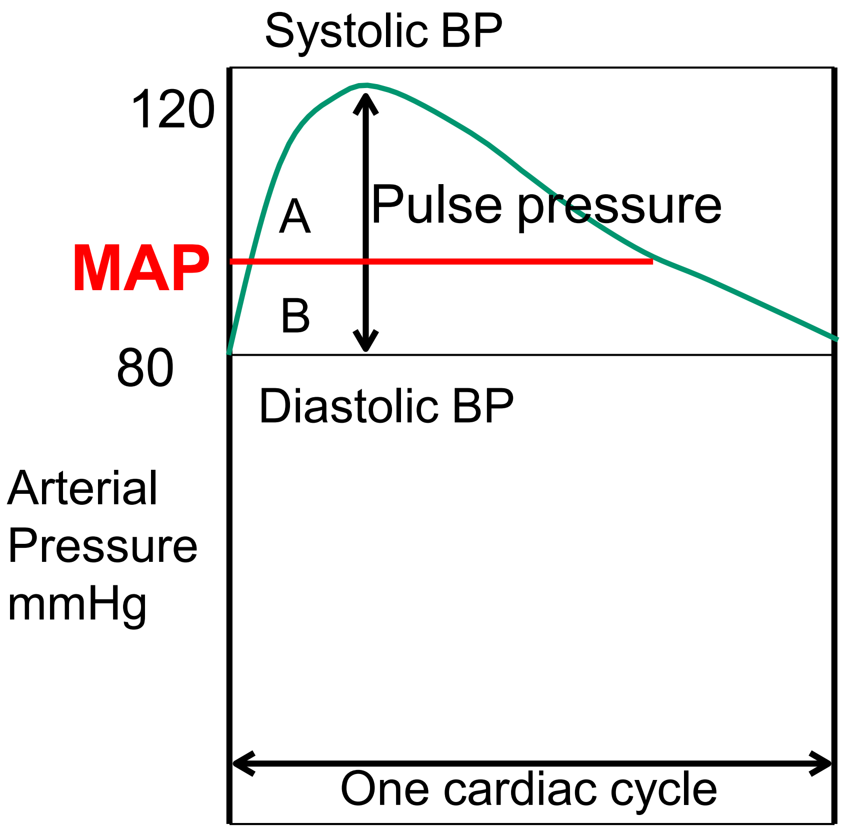


- During systole: The ventricle ejects blood into aorta faster than it can flow away into the periphery. This causes a steep rise in aortic pressure.
- ~70% of the stroke volume is stored initially in the elastic arteries during systole. Remaining 30% runs off through the peripheral vessels.
- Max. pressure in the aorta is the Systolic BP=120 mmHg
- As blood runs into the peripheral circulation the aortic BP falls.
- Minimum pressure in aorta is the Diastolic = 80 mmHg

- Therefore, arterial compliance (systolic BP) and peripheral resistance (diastolic BP) determines **stroke volume** – pulse pressure.

Pulse Pressure = SBP – DBP = 120 – 80 = 40 mmHg

Arterial pressure wave:
Blood Pressure in the Aorta during one cardiac cycle



- What is the average blood pressure in the aorta to maintains good blood flow and tissue perfusion?

- The MAP is the mean arterial pressure in the aorta during a single cardiac cycle.
- This setpoint is determined by the cardiac output and total peripheral resistance.

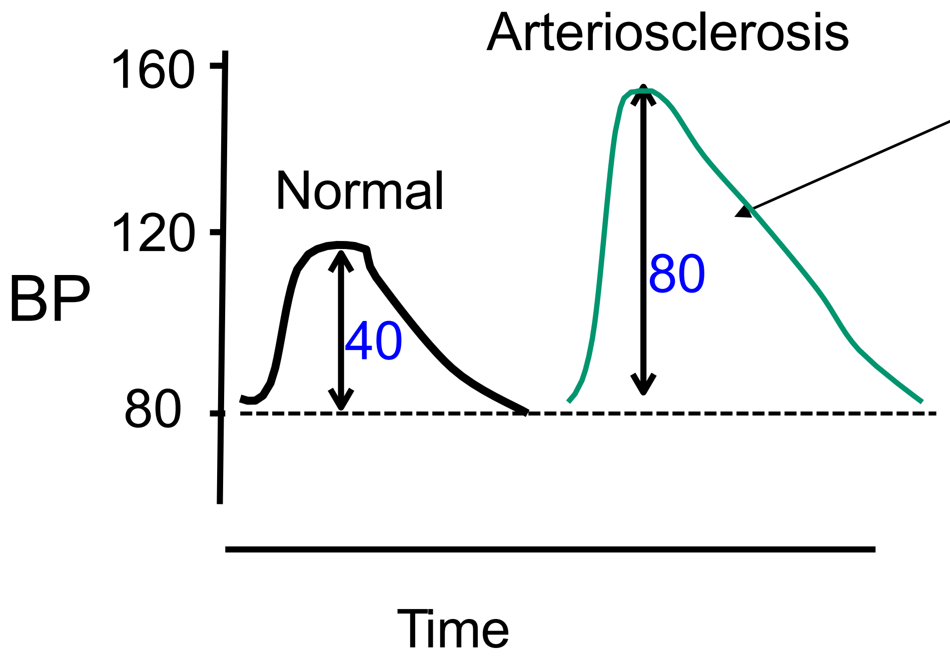
- On the graph, at the line where areaA = areaB.
- Value is usually ~ 95 mmHg

MAP would be more beneficial and accurate in identifying the cerebrovascular impact of hypertension,
PP predictor of cardiac events

MAP = $P_{dias} + \frac{1}{3}(P_{sys} - P_{dias}) = 80 + 1/3(40) = \sim 93$ mmHg

Blood pressure and ageing
Factors that affect arterial pressure waves, MAP and PP

1. Stroke Volume: *SV α Pulse Pressure
2. Arterial compliance: Compliance α 1/pulse pressure With increasing age, compliance decreases and pulse pressure increases.



Aorta becomes stiffer and less compliant in old age.

Elderly: BP 160/80 mmHg

"Systolic hypertension of the elderly"

Don't confuse arteriosclerosis and atherosclerosis. Arteriosclerosis is stiffening of the arteries - a consequence of ageing. Atherosclerosis is plaque formation.

*This is a simplification, because we are assuming that none of the SV is leaving the aorta

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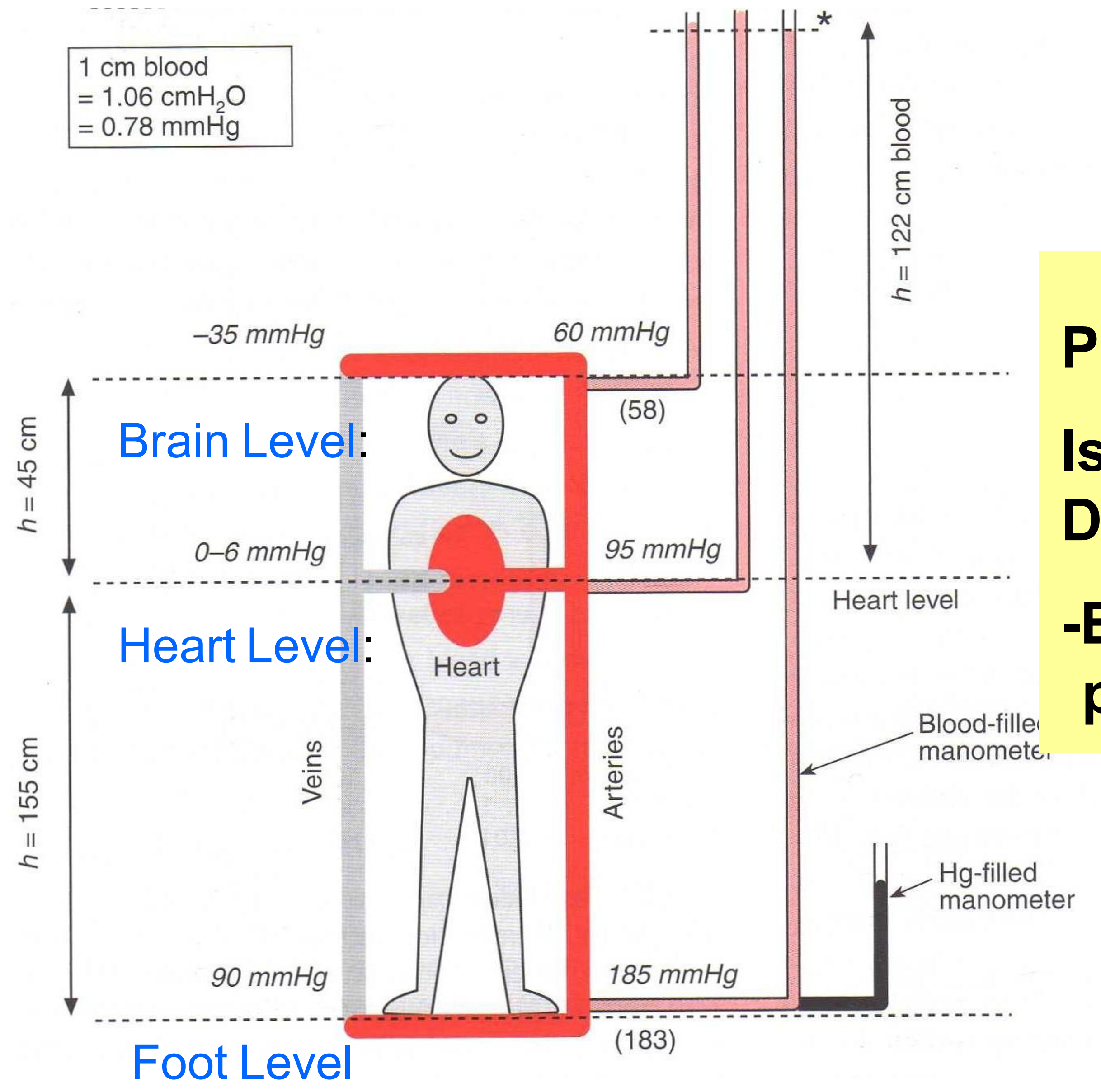
Effects of gravity on blood pressure



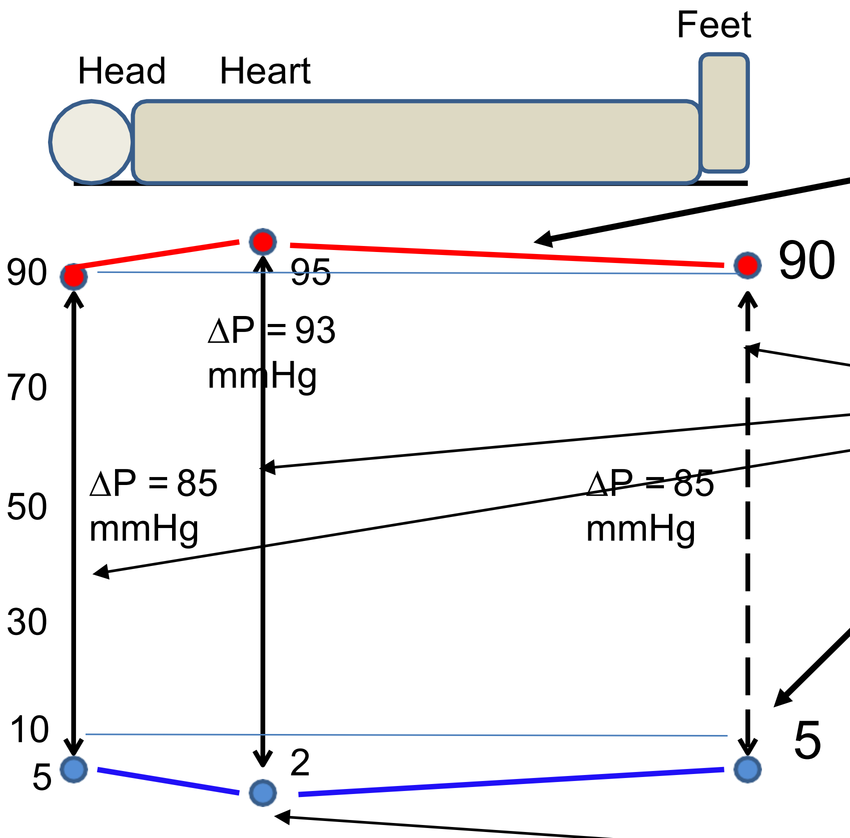
Pressure Foot > heart

Is this in defiance of Darcy's law Q α ∆P?

-Blood flows against pressure gradient

The Effect of Posture on Arterial and Venous Pressures

In the Supine Position – the entire body is at the level of the heart.



Pressure drop of ~ 5 mmHg between aorta and dorsalis pedis artery in foot. Pressure in foot = 90 mm Hg

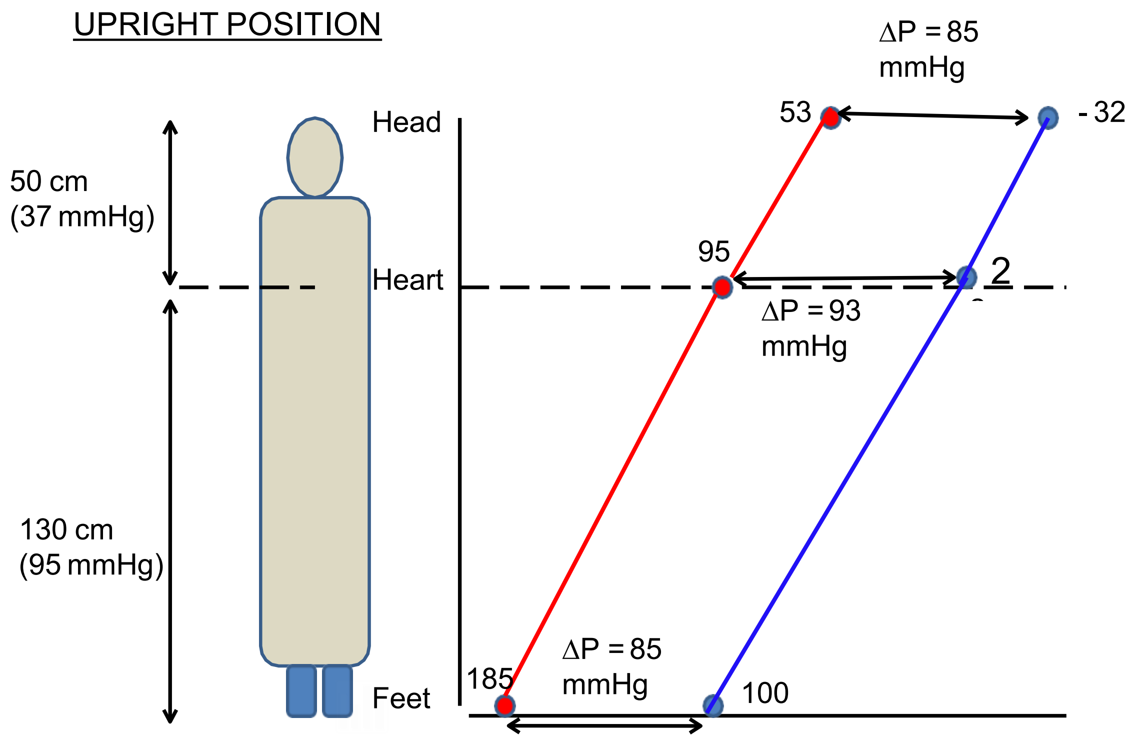
ΔP = Driving Pressure

Pressure in veins draining feet is 5 mm Hg.

Pressure falls to 2 mm Hg at RA

The Effect of Posture on Arterial and Venous Pressures

On Standing:



Gravity causes a volume of blood to pool in the lower limbs.

1. Add the weight of the column of blood (hydrostatic pressure) to BOTH the arterial supply of the feet AND to the veins draining the feet. (130cm=95mmHg)

- Arterial pr (in foot) = 90 **+ 95** = 185 mmHg
  Venous pr (foot) = 5 **+ 95** = 100

2. Subtract the weight of the column of blood (hydrostatic pressure) from BOTH the arterial supply of the head AND from the veins draining the head. (50cm=37mmHg)

- Arterial pr (in head) = 90 **- 37** = 53 mmHg
  Venous pr (in head) = 5 **- 37** = -32 mmHg

∴ Driving pressure = 185 – 100 = 85 and 53 - -32 = 85mmHg (same as in supine person)

As long as there is a pressure difference between arteries and veins – blood will flow!

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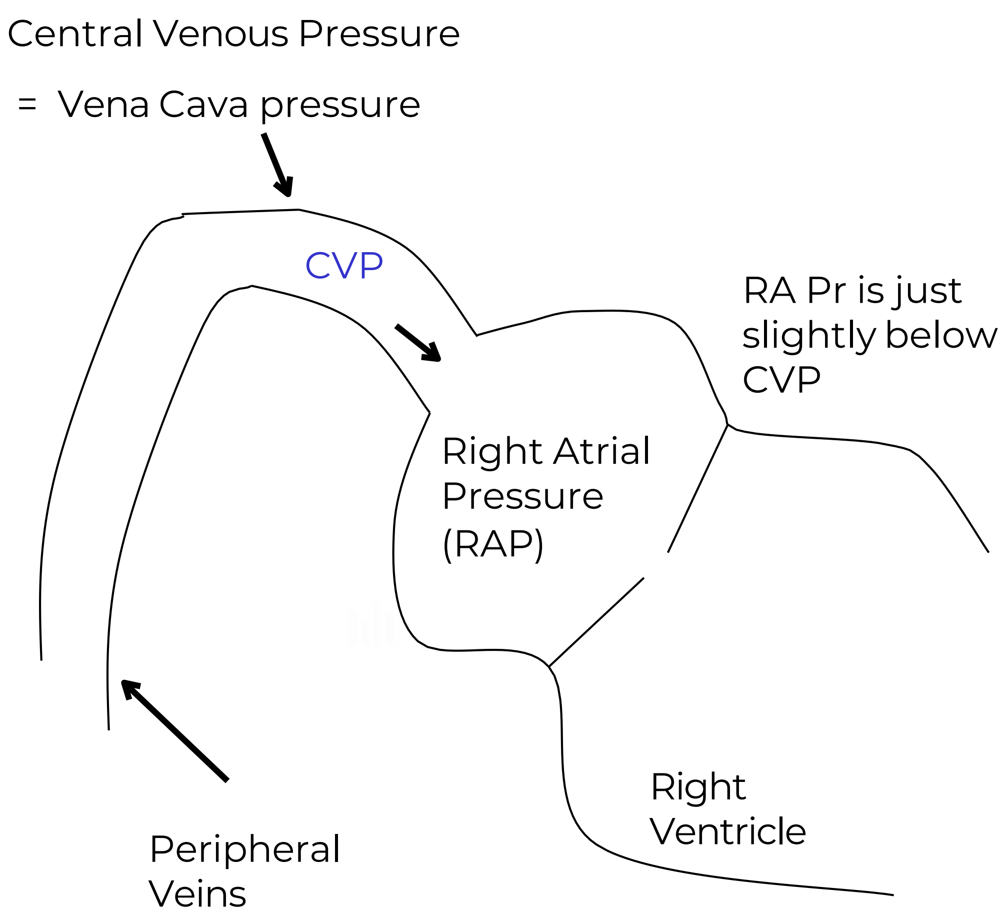
Venous pressure, central venous pressure and right atrial pressure

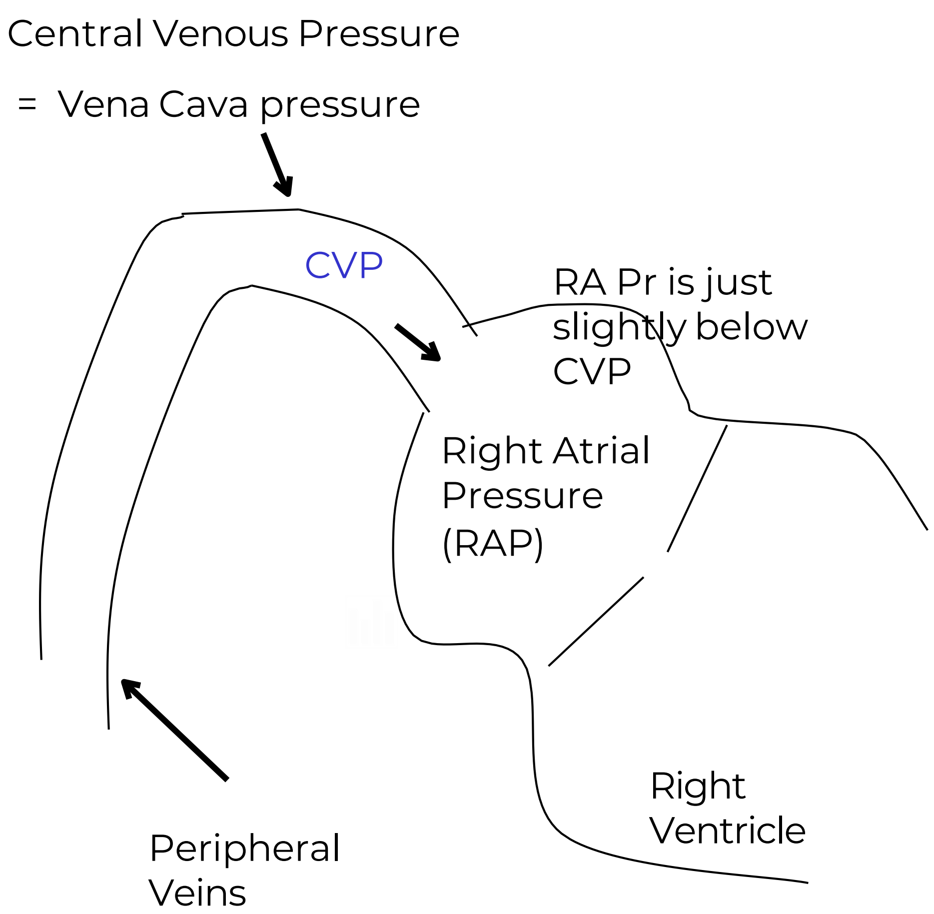


1. Venous pressure - pressure in the venous compartment, altered by changes in venous tone

2. Central venous pressure – pressure in thoracic vena cava, just before emptying into right atrium. Important - it determines, filling pressure, pressure gradient, venous return and stroke volume /preload via Frank-Starling mechanism

3. Right atrial pressure – blood pressure in the right atrium of the heart. Nearly identical to CVP. RAP depends on balance between ability of heart to pump blood out into circulation and tendency of blood to flow back into heart

Venous pressure, central venous pressure and right atrial pressure



1. **Right atrial pressure** – almost identical to CVP

Determines preload – stroke volume and venous return

Factors that alter RAP are:

I. **RV function**
   I. RV contraction – strong = ↓RAP (norepinephrine)
   II. RV contraction – weak = ↑ RAP (heart failure)

Cardiac Failure
RAP increases (20 – 30 mm Hg). CVP increases, capillary hydrostatic pressure increases. Favouring edema formation.

II. **↑ Venous return** = ↑ RAP
   I. ↑Blood volume (IV infusion)
   II. ↑Venomotor tone (venoconstriction)

Sympathetically-induced venoconstriction displaces blood from the peripheral veins into central veins and into RA, thus ↑ VR.

III. **↓Venous return** = ↓ RAP
   I. ↓ Blood volume – hemorrhage
   II. ↓ Venomotor tone (venodilation)

Mechanical factors affecting: Venous return

Venous pressure, central venous pressure and right atrial pressure

MUSCLE PUMP

Before muscle contraction blood enters vein

→

contraction of skeletal muscles

>compress veins rhythmically >displaces blood forward toward heart

→

valves in veins prevent backflow of blood

→

CVP is maintained, maintaining VR and EDV

RESPIRATORY PUMP

During Inspiration: VR ↑ to RA

→

On Inspiration

↓ Intrathoracic Pressure >Thoracic veins expand

Contraction of diaphragm>↑ abdominal pressure>↑

→

Blood from head fills the vena cava> ↑ Flow into vena cava

→

↑ Venous Return*

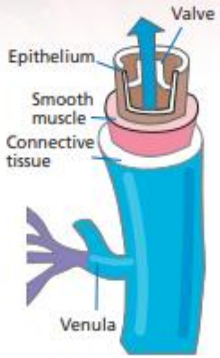
*physiological splitting of S2 during deep inspiration

Venous return mechanisms

Venous return is the return of blood to the right side of the heart via the vena cava. Up to 70% of the total volume of blood is contained in the veins at rest. This means that a large amount of blood can be returned to the heart when needed. During exercise, the amount of blood returning to the heart (venous return) increases. This means that if more blood is being pumped back to the heart then more blood has to be pumped out, so stroke volume will increase

The pressure of the blood in the large veins is very low and this makes it difficult to return blood back to the heart. In addition the large lumen of the vein offers little resistance to blood flow.

This means that **active mechanisms** are needed to help venous return.

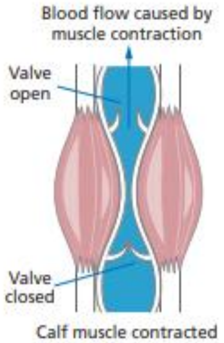


Smooth muscle

A thin layer of smooth muscle in the veins helps squeeze blood back towards the heart.

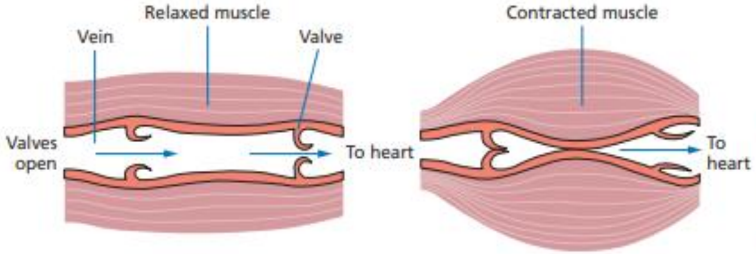
Valves

Valves prevent the backflow of blood.



Skeletal muscle pump

When muscles contract and relax, they press on nearby veins, causing a pumping effect and squeezing the blood towards the heart.

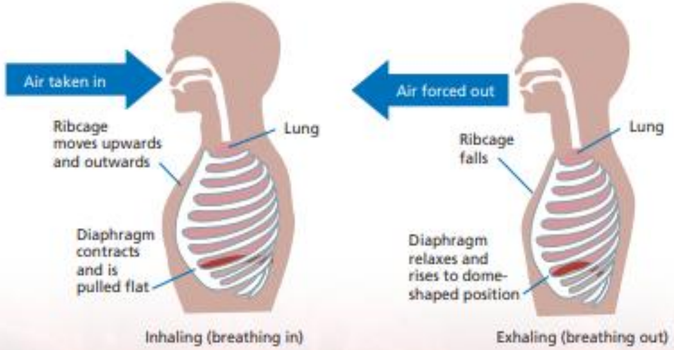


Gravity

Gravity helps the blood return to the heart from the upper body.

Respiratory pump

When muscles contract and relax during the inspiration and expiration process, pressure changes occur in the thoracic and abdominal cavities. These pressure changes compress the nearby veins and assist blood return to the heart.



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Lecture Outline

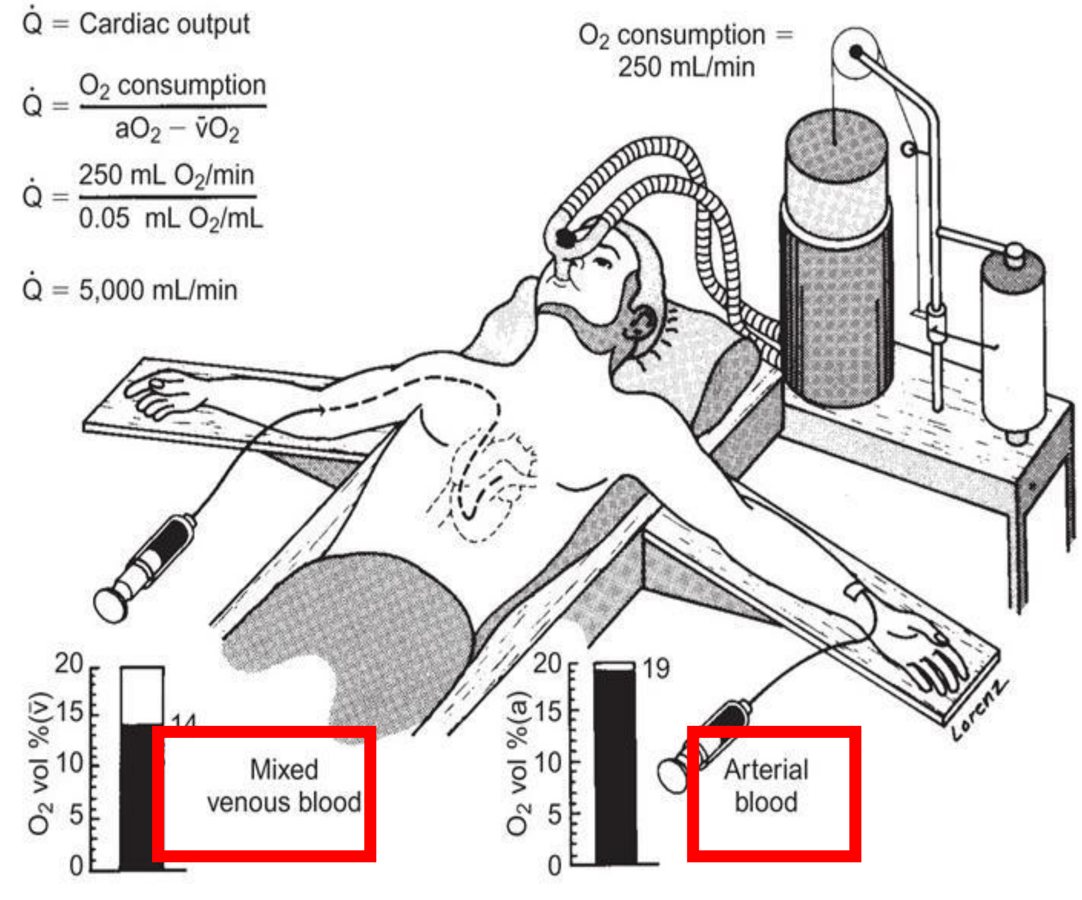
- Types of blood flow
  - Laminar vs turbulent flow,
  - Reynolds equation,
  - Single line flow
- Physiological properties of vessels
  - La Place's Law of wall tension
  - Compliance vs elastance
- Pulse Pressure and Mean Arterial Pressure
- Effect of gravity on blood pressure
- Factors affecting venous return
- **Measurement of Cardiac Output: Fick's principle**

Measuring blood flow – Fick principle

Oxygen extraction

Law of conservation of mass:

Rate O_2 removal = $[O_2]$ pulmonary flow in (artery) - $[O_2]$ pulmonary flow out (veins)



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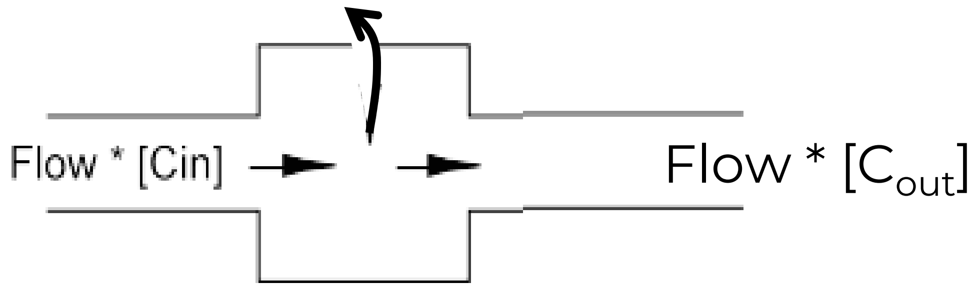
Rate O_2 removal **(over time)**



- Rate of O_2 **loss**/min
- Amt of O_2 loss/vol
- Can calc blood flow→ ml/min

100 ml blood = 19 mls O_2 arterial (0.19 per ml)
100 ml blood = 14 mls O_2 venous (0.14 per ml)
i.e. amt of O_2 loss per ml blood gave 0.05 mls O_2. Know O_2 extraction/time **250**
What vol of blood holds 250 mls O_2 every min = 5000 ml

Hemodynamic principles summary

- Blood flow = Pressure gradient/resistance; **FLOW(Q) = ΔP / R**
- Blood flow similar to Ohm's law = I=V/R

Darcy'sLaw

- Resistance = **ΔP**/Q;
- Resistance α $\frac{8 \text{ x viscosity x length}}{\pi \text{ x radius }^4}$ = $\Delta P \text{ x } \frac{\pi r^4}{8\eta L}$

Poiseuille's Law

- **Flow α radius ^{4}**
- SERIES: **$(R_{total}) = [\ R_1 + R_2 + R_3 + R_4\ ]$**
- PARALLEL: **$(R_{total}) = 1/\ [R_1 + R_2 + R_3 + R_4\ ]$**
- Velocity = Flow / Cross sectional Area
- Reynold's number(2000) = $v\ D\ \rho\ /\eta$
- La Place's Law for a blood vessel: $\Delta P = T/r$
- Cardiac output = O2 consumption /$[O_2\ pv] - [O_2\ pa]$
- CO = MAP-RAP/TPR
- CO=SV*HR

High flow velocity v

Large vessel diameter D

High blood density ρ

Viscosity η


Photo credits: Dr. GWB